Seat No.: _____ Enrolment No._____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–IV (NEW) EXAMINATION – SUMMER 2022

Subject Code:3144202 **Date:02-07-2022**

Subject Name:Discrete Mathematics and Graph Theory

Time:10:30 AM TO 01:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Q.1 **(a)** If p stands for 'it is cold' and q stands for 'it is raining', translate the following into symbols **03**

(i) It is cold or it is raining

(ii) It is cold and it is raining

(iii) A necessary and sufficient condition for raining is that it be cold

(b) Let $f: R \to R,\ \ f(x) = 5x + 4$ and $g: R \to R,\ \ g(x) = 4x + k$ **04**

If $fog = gof$ then find k.

(c) Using Venn diagrams, verify the following identities considering sets **07**

$U = \{1,2,3,4\}$,

$X = \{1,2,3\}$ and

$Y = \{2,3,4\}$.

(1) $Y - (Y - X) = X$

(2) $X - (Y \cap X) = Y - X$

Q.2 **(a)** Let A be the set of all straight lines in a plane. **03**

Let R = {(x, y) : x, y ∈ A and x ⊥y}

Examine if R is reflexive, symmetric and transitive.

(b) Construct the truth table and verify logical equivalence **04**

$\sim (p \vee q) \equiv \sim p \wedge \sim q$

(c) Consider these statements. **07**

"All lions are fierce."

"Some lions do not drink coffee."

"Some fierce creatures do not drink coffee."

Let P (x), Q(x), and R(x) be the statements

"x is a lion," "x is fierce," and "x drinks coffee," respectively.

Assuming that the domain consists of all creatures, express the statements in the argument using quantifiers and P (x), Q(x), and R(x)

OR

(c) Consider these statements, **07**

"All hummingbirds are richly colored."

"No large birds live on honey."

"Birds that do not live on honey are dull in color."

"Hummingbirds are small."

Let P (x), Q(x), R(x), and S(x) be the statements

"x is a hummingbird," "x is large," "x lives on honey," and "x is richly colored," respectively.

Assuming that the domain consists of all birds, express the statements in the argument using quantifiers and P (x), Q(x), R(x), and S(x).

Q.3 **(a)** Let $A = \{1,2,3\}$ be a given finite set and **03**
$P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{3,1\}, \{1,2,3\}\}$ be its power set.
Let $\subseteq$ be the inclusion relation on the elements of $P(A)$.
Draw Hasse diagram for $\langle P(A), \subseteq \rangle$. Is $\langle P(A), \subseteq \rangle$ a lattice ? Justify.

(b) Draw the Hasse diagrams of the following sets under the partial ordering relation "divides" and indicate those which are totally ordered. **04**
(i) $\{1,2,3,4\}$ (ii) $\{3,5,15\}$ (iii) $\{2,4,8,16\}$ (iv) $\{1,2,3,6,12\}$

(c) Let Z be the set of integers and R be the relation called "Congruence modulo 5" defined by R = $\{\langle x, y\rangle \mid (x - y)$ is divisible by 5$\}$ **07**
Show that R is an equivalence relation. Determine the equivalence classes generated by the elements of Z

OR

Q.3 **(a)** Draw directed-graph from the given adjacency matrix **03**

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

(b) Solve the recurrence relation $a_n - 3a_{n-1} = 2\,; n \geq 1\ with\ a_0 = 1$ using the method of undetermined coefficients. **04**

(c) Find maximal compatibility blocks of the relations shown in following figures. Also write their relation matrices. **07**

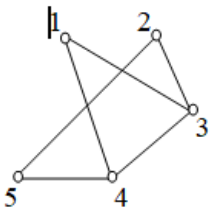


Figure-1

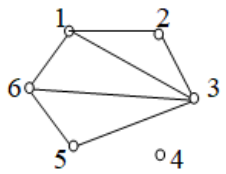
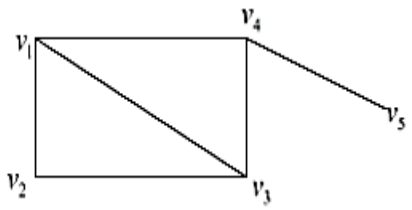
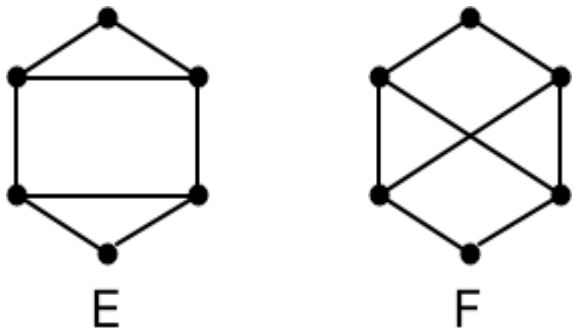


Figure-2

Q.4 **(a)** Find the adjacency matrix for the given graph **03**



(b) Are the following graphs isomorphic? Justify your answer. **04**

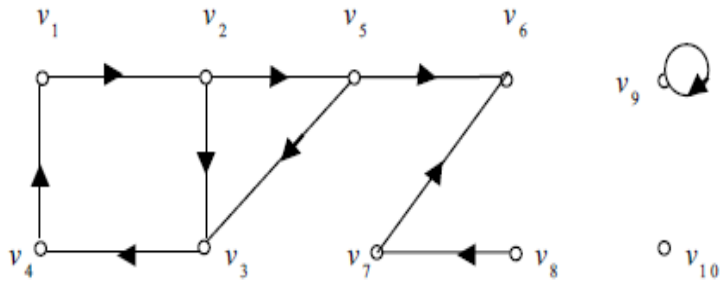


(c) Explain the number of vertices and number of edges of following by drawing graphs: (1)K_3 (2)$K_{2,2}$ **07**

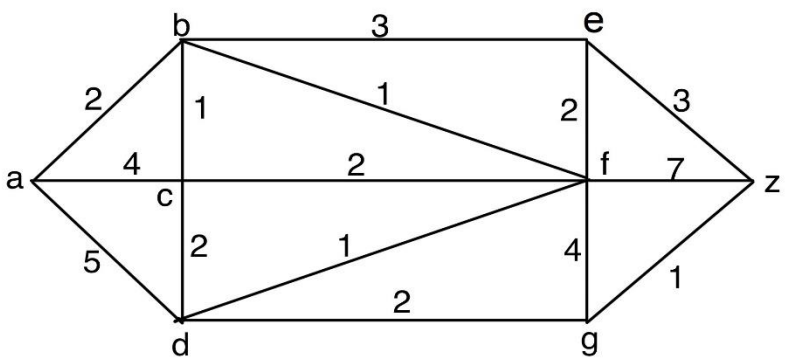
OR

Q.4 **(a)** Define Graph, Diagraph, Mixed graph with suitable example **03**

(b) Find the reachable sets of node v_1 and node v_8 for the digraph given in following Figure. **04**



(c) Using Dijkstra's algorithm, find the shortest path between vertex a and z **07**



Q.5 **(a)** Let $R = \{0°, 60°, 120°, 180°, 240°, 300°\}$ **03**
Draw the composite table for the operation * defined by
$a * b = Over\ all\ a$ngular rotation corresponding to successive rotations by a and then by b , $\forall a, b \in R$,

(b) Check the set $G = \{a + \sqrt{2}b : a, b \in Q\}$ is a group with respect to addition.. **04**

(c) Show that $(Z_4,+_4)$ is a cyclic group and find all the subgroups of $(Z_4,+_4)$. **07**

OR

Q.5 **(a)** Define an abelian group. Show that if every element in a group is its own inverse, then the group must be abelian. **03**

(b) Consider a set of integers 1 to 500. Find how many of these numbers are divisible by 3 or 5 or by 11? **04**

(c) Let $(H_1,*)$ and $(H_2,*)$ be subgroups of a group $(G,*)$. Show that $(H_1 \cap H_2,*)$ is also a subgroup of $(G,*)$. Will $(H_1 \cup H_2,*)$be also a subgroup of $(G,*)$? **07**

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